

Console Tester lessons learned

This postmortem records the failures that materially shaped Public Betas 0.7 and 0.8. It is written for maintainers who need to understand why apparently simpler changes can break real recovery hardware.

A diagnostic cannot depend on the part it diagnoses

The first keyboard page used the console ROM keyboard service. It worked on a healthy /4A but failed the intended recovery case: a machine with missing or damaged GROMs could start through LOAD and then lose input. The final firmware owns the TMS9901 scan, translation, modifiers, Alpha Lock, joysticks, debounce, VDP access, PSG output, and navigation. The build rejects any executable call to the console keyboard vector.

Firmware identity follows the same principle. Stable replacement ROM/GROM data is descriptive, not a failure. Only an unstable read path is diagnostic.

The /4 and /4A are different keyboards

The 40-key /4 is not a shortened /4A matrix. It uses five keyboard columns and SHIFT for the printed direction/function overlay. The /4A adds its own sixth column, FCTN, CTRL, Alpha Lock, and two unpopulated contacts. Separate fixed tables prevent plausible-looking but incorrect translations.

Modifier state and the translated character are separate facts. Treating every Control chord as CTRL-A, or inferring Alpha Lock from a joystick row, produced misleading displays. The final page reports the raw chord and readable result, including direction names, FCTN punctuation, and AUX-1/AUX-2.

Expose uncertain wiring before naming it

Surviving diagrams disagreed about the TI-99/4 joystick selectors. A temporary diagnostic displayed all candidate columns; physical testing established joystick 1 on column 5 and joystick 2 on column 6. The /4A family remains on columns 6 and 7. Raw evidence settled the question before the friendly labels were frozen.

A VDP page must own all VDP state

The VDP demo changes the name, pattern, color, and sprite table selections. An early return path redrew text without first restoring every register, so later pages such as Credits and the VRAM chip map inherited demo colors, sprites, or tables. The text had been written to the intended VRAM addresses; the VDP was simply fetching a different layout.

The correction established one common boundary: blank, program the UI state, rebuild the font/colors/frame/sprite terminator, reassert all eight registers, then reveal the page. The regression is tested as a sequence of pages after the VDP demo, not as isolated screenshots.

Timing-sensitive code needs its scheduler

The MegaDemo raster effect appeared slow, turbo-fast, jagged, silent, or frozen when its inner loop was called without the original frame boundary. Software delays could slow music while leaving scanline work badly aligned. The working adapter waits for a fresh VDP interrupt, preserves the caller's return link, and calls the original raster controller exactly once per display frame. The approved PSG bytes remain unchanged.

The lesson is broader than this demo: transplant timing code with its event source, state, workspace, and return convention. Do not tune unrelated delays until two broken clocks happen to look close.

Runtime placement is part of compatibility

An early demo build ran from optional expansion RAM at `>A000`. That worked in a development environment but froze on the standalone tester, which cannot assume a 32 KiB expansion and conflicts with a normal expansion at higher addresses. The final controller is relocated as a unit to tester U7 SRAM at `>D000`.

Sprite status is not visible sprite count

The fifth-sprite status flag reports a scanline limit; it does not say that a fifth object was displayed. Overlapping entries can also make several sprites look like one. In the first test layout, a white high-priority sprite sat exactly over a red sprite on a white background, hiding the collision pair. Beta 0.8 instead puts the red sprite first and exactly overlaps the white one, so the pair appears as one clean red object while preserving collision. The yellow fifth entry should remain hidden at the stock limit, but may appear on replacement hardware configured for a higher scanline limit.

The UI reports collision and fifth-sprite flags as `SET` or `NOT SET` and asks the operator to inspect shapes, sizes, and colors. It does not turn a replacement VDP's configured sprite limit into a false hardware failure.

Reset every sprite table a demo owns

Restoring the ordinary UI sprite table at `>0300` was not sufficient after the MegaDemo. Its raster controller uses a separate timing-sprite table at `>3800`. On a real TMS9918A, those timing sprites could remain visible over the results, VRAM map, credits, and later menus even though the text palette and border had been restored. The text initializer now blanks the display, erases both full 128-byte tables, restores all legacy VDP registers, and terminates both tables again before enabling the display.

A helper's register side effects are part of its interface

The first enhanced-VDP detector appeared correct in source but never unlocked an F18A. Its VDP-register helper byte-swapped `R0` while formatting the control word. Calling the helper twice after loading register 57 only once therefore sent two different transactions instead of the required identical `>1C` writes.

Beta 0.8 emits both complete unlock transactions explicitly and protects their machine-code sequence in the static tests. When hardware protocols require repeated writes, reload the value or use a routine whose input-preservation contract is explicit; visual similarity at the call site is not proof of identical bus traffic.

Freeze bytes, not filenames

Repeatedly reusing a development filename made it too easy to test the wrong image. The public build now has a TI-compatible release alias, a manifest, SHA-256 sums, and a source snapshot. Comment-only cleanup is accepted only when the resulting programmer image remains byte-identical to the validated build.

The public repository preserves released betas but excludes intermediate test images, disassemblies, ROM dumps, emulator loaders, and private harnesses. Those materials are useful for development archaeology but make a poor build contract for new contributors.